Groups and Lie Algebras: Exercise Sheet 3

1. **Some properties of characters**

   Let V and W be a finite dimensional representation of a finite group G with characters χ_V and χ_W. Compute the character of the representations $V \oplus W$, $V \otimes W$, V^*, $\mathrm{Hom}(V, W)$, $\Lambda^2 V$, and $\mathrm{Sym}^2 V$.

2. **Quaternions**

   The quaternion group consists of 8 elements $\pm 1, \pm i, \pm j, \pm k$. The defining relations are

   $$i^2 = j^2 = k^2 = -1, \quad i = jk = -kj, \quad j = ki = -ik, \quad k = ij = -ji$$

   a) First find 4 one-dimensional representations of Q_8.

   b) There is also a 2-dimensional irreducible representation. To find it, think of quantum mechanics.

   c) Are these all irreducible representations? Explain.

   d) Compute the character table.

3. **The symmetric group S_4 and its representations**

   a) Determine the conjugacy classes of S_4 and associate Young diagrams and conjugacy classes.

   b) There are two 1-dimensional representations of S_4. What are they?

   c) Consider the 4 dimensional standard representation of S_4 on $\mathbb{R}^4$. Decompose it into a 3 dimensional irreducible representation and a 1 dimensional representation.

   d) Take the tensor product of the 3 dimensional representation of the previous part and the alternating representation. It is again irreducible. How many other irreducible representations are there? What are their dimensions?

   e) Write down the character table.

4. **Particle in a box**

   The Hilbert space of a particle in a box of length l is $L^2([-l, l]^2)$ with the boundary conditions $\psi|_{\partial[-l,l]^2} = 0$ and Hamiltonian $\mathbb{H} = -\frac{\hbar^2}{2m}(\partial_x^2 + \partial_y^2)$. Show that this system has D_4 symmetry. Solve the Schrödinger equation and find an energy level whose eigenfunctions transform in the 2-dimensional representation of D_4. What is the lowest energy level that does not transform in an irreducible representation of D_4.